

Superstore Sales Performance and Profitability Analysis

Sector: Retail / E-Commerce



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Problem

The retail business generates substantial revenue (\$3.48M), yet overall profitability remains low (~6.35% margin). Analysis reveals that:

- High discounts significantly erode profit.
- Certain categories generate high sales but weak returns.
- Regional performance varies widely.
- Revenue-based decision-making hides loss-making segments.

Without structured analytics, management risks prioritizing volume over profitability.

Approach

To address this issue, a structured Retail Performance Analytics model was developed using transactional sales data.

The methodology included:

- Data cleaning and transformation in Google Sheets.
- Creation of analytical KPIs:
 - Profit Margin
 - Discount Bands
 - Product Strategic Segments
- Regional, category, and discount impact analysis.
- Trend analysis across months.
- Operational review of shipping efficiency.
- Dashboard visualization for executive interpretation.

All calculations and pivots were executed manually in Google Sheets as per capstone requirements.

Key Insights

- High discounts (>20%) significantly reduce profitability.
- The technology category delivers the highest margins.
- Furniture generates revenue but weak profit margins.
- The West region contributes the highest profit share.
- The South region underperforms in both sales and margins.
- A small group of products drives most of the total profit.
- Sales show seasonal peaks in September and November.

Recommendations

- Cap discounts at 15–20% to protect margins.
- Increase investment in Technology products.
- Reprice or optimize Furniture category.
- Improve pricing and promotional strategy in the South region.
- Shift performance tracking from revenue to profit margin.
- Focus marketing on high-margin “Star” products.

Sector Overview

This project belongs to the Retail & E-Commerce Analytics sector.

Retail operations involve high transaction volumes, multiple product categories, regional distribution, and discount-driven competition.

Profitability management is a key challenge in this sector due to:

- Competitive pricing pressure
- Promotional discounting
- Inventory complexity
- Regional demand differences

Current Challenges in Retail

- Margin compression due to discount wars
- Revenue-focused performance tracking
- Lack of product segmentation strategy
- Regional inefficiencies
- Operational cost variability (shipping, logistics)

Why Was This Problem Chosen?

Many retail firms mistakenly prioritise revenue growth while ignoring profitability erosion.

This project was chosen to demonstrate how:

-
- Data analytics reveals hidden losses
 - Discount strategy impacts bottom line
 - Product segmentation improves decision-making
 - Profit-focused dashboards drive better strategy

Formal Problem Definition

To develop a data-driven analytical framework that evaluates sales transactions to identify profitability drivers, discount impact, regional performance disparities, and product segmentation efficiency.

Project Scope

Includes:

- Sales and profit analysis
- Discount impact assessment
- Regional comparison
- Category-level profitability
- Product segmentation model
- Shipping efficiency analysis

Excluded:

- Predictive forecasting
- Machine learning modeling
- Real-time data integration

Success Criteria

The project is successful if it:

- Identifies loss-making discount bands
- Highlights high-margin categories
- Detects weak-performing regions
- Classifies products strategically
- Provides actionable recommendations

Data Description

Dataset: Superstore Retail Dataset

Source: Kaggle

Access Link: [Superstore Sales: The Data Quality Challenge](#)

Data Structure

- Transaction-level dataset
- Each row represents a product line item within an order
- Multi-year sales data
- Multiple regions and categories

Data Size

- ~12,000+ transaction records
- \$3.48M total sales
- \$228K total profit
- 4 regions
- 3 primary categories

Data Limitations

- No cost breakdown per product
- No marketing campaign data
- No customer demographic data
- Historical dataset (not real-time)
- External economic factors not included

Derived Columns Created:

- Profit Margin %
- Discount Band
- Product Strategic Segment
- Month & Year Extracted

Columns Explanation

Column Name	Description
Order Date	Date on which the customer placed the order.
Ship Date	Date on which the order was shipped to the customer.
Region	Sales region where the order was processed (e.g., West, East, Central, South).
State	State corresponding to the customer's shipping location.
Category	High-level product classification (e.g., Technology, Furniture, Office Supplies).
Sub-Category	Detailed product classification within each main category.
Product Name	Name or description of the individual product sold.
Sales	Total revenue generated from the transaction (after discount).
Quantity	Number of units sold in the transaction.
Discount	Percentage discount applied to the product.
Profit	Net profit earned from the transaction after applying discount and costs.

Data Cleaning & Preparation

All data cleaning and transformation were executed in Google Sheets as per capstone requirements.

Missing Values Handling

- Blank discounts replaced with 0.
- Date formatting standardized.
- Removed inconsistent text formatting.
- Verified no null sales or profit records.

Outlier Treatment

- High discount values retained (important for analysis).
- Negative profits retained (critical for business insight).
- Extreme values verified against dataset consistency.

No artificial trimming was performed to preserve business realism.

Transformations

- Created Month column from Order Date.
- Calculated Profit Margin:

Profit / Sales

- Converted discount into categorical bands:
 - 0–10%
 - 10–20%
 - 20–30%
 - 30–40%
 - 40%+
- Aggregated data using Pivot Tables.

Feature Engineering

1. Profit Margin %
2. Discount Band Classification
3. Product Strategic Segment:
 - Star Product
 - Hidden Gem
 - Loss Maker
 - Low Performer
 - Volume Driver
4. Regional Contribution %
5. Category-Level Margin

Assumptions

- Profit values reflect true net profitability.
- Discounts directly influence margin.
- Each row represents one valid transaction.
- Dataset is representative of retail business operations.

KPI & Metric Framework

To evaluate retail performance and profitability trends, five core Key Performance Indicators (KPIs) were designed. These KPIs align directly with the project objective of shifting focus from revenue growth to sustainable profit optimization.

Core KPIs

1. Total Sales

Definition: Total revenue generated from all transactions.

Formula:

Total Sales = Σ Sales

Purpose: Measures overall business scale and revenue growth.

2. Total Profit

Definition: Net earnings generated after discounts and transaction costs.

Formula:

Total Profit = Σ Profit

Purpose: Reflects actual financial performance and sustainability.

3. Total Orders

Definition: Total number of completed transactions.

Formula:

Total Orders = Count of Order ID

Purpose: Indicates demand volume and transaction frequency.

4. Profit Margin (%)

Definition: Profit earned as a percentage of total sales.

Formula:

Profit Margin (%) = $(\text{Total Profit} / \text{Total Sales}) \times 100$

Purpose: Standardizes profitability comparison across time, region, and categories.

5. Year-over-Year (YoY) Growth (%)

Definition: Percentage change in KPI performance compared to the previous year.

Formula:

$$\text{YoY Growth (\%)} = ((\text{Current Year} - \text{Previous Year}) / \text{Previous Year}) \times 100$$

Purpose: Evaluates sustainability of revenue and profit growth trends.

KPI Alignment with Project Objectives

Project Objective	Supporting KPIs
Analyze revenue vs profitability gap	Total Sales, Total Profit
Detect margin erosion	Profit Margin
Evaluate growth sustainability	YoY Growth
Assess demand volume impact	Total Orders
Shift focus from revenue to profit intelligence	Profit Margin, Total Profit

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted on 12,617 cleaned transactions (2014–2017) to identify patterns, relationships, and profitability drivers. The analysis focuses on trends, comparisons, distributions, and correlations to support data-driven decision-making.

Trend Analysis

Trend analysis was performed using monthly and yearly aggregations of Sales and Profit.

Key Findings:

- Sales peaked in **September (\$462K)** and **November (\$444K)**.
- Year-end months consistently show stronger performance compared to mid-year periods.
- A moderate slowdown is observed during certain mid-year months.
- Profit growth does not always increase proportionally with sales due to discount effects.

Insight:

The business demonstrates clear seasonal demand patterns. Promotional campaigns and inventory planning should be aligned with high-performing months to maximize profitability rather than just revenue.

Chart Used: Monthly Sales Trend Line Chart

Comparison Analysis

Regional Comparison

- East and Central regions show moderate performance.
- The **West region contributes 37.2% of total profit**, making it the strongest-performing region.
- The **South region shows weaker profitability** (Sales: \$646K, Profit: \$42K).
- East and Central regions show moderate performance levels.

Insight:

Regional imbalance suggests differences in pricing efficiency, demand dynamics, or operational performance. The West region can be used as a performance benchmark.

Chart Used: Regional Sales vs Profit Bar Chart

Category Comparison

- **Technology** has the highest profit margin (~9.9%).
- **Office Supplies** maintains moderate margins (~5–6%).
- **Furniture** has the weakest margin (~0.74%) with 44% of products unprofitable.

Insight:

High revenue does not guarantee high profitability. The Furniture category requires pricing revision and product rationalization.

Chart Used: Category-wise Sales and Profit Comparison Chart

Distribution Analysis

Distribution analysis was performed using discount bands and profit margins.

Observations:

- Transactions with **0–10% discount** generate strong positive margins.
- Profit margins decline sharply as discount levels increase.
- When discounts exceed **20%**, profitability becomes unstable or negative.
- Approximately **23% of transactions have discounts above 20%**, contributing significantly to margin erosion.

Insight:

A clear discount threshold exists beyond which profitability deteriorates. Discount strategies should be carefully controlled to protect margins.

Chart Used: Discount Band vs Average Profit Margin Line Chart

Correlation Analysis

Correlation analysis was conducted between Discount, Sales, and Profit Margin.

Key Relationships:

- **Strong negative relationship between Discount and Profit Margin.**
- Higher discounts slightly increase sales volume but significantly reduce margin.
- Product-level analysis shows high profit concentration — top-performing products contribute a disproportionate share of overall profit.

Insight:

Discount-driven revenue growth is not sustainable in the long term. A profit-focused strategy should prioritize high-margin

Analysis Source: Discount Band Pivot Table

Consolidated Insights from Visualizations

- Sales exhibit month-level performance variation.
- Profitability varies significantly across regions and categories.
- Discounting is a primary driver of margin erosion.
- Profit contribution is unevenly distributed across products.
- Re-focused decisions may conceal underlying profitability risks.

Advanced Analysis

Segmentation Analysis

Product strategic segmentation performed:

- Star Products → Core revenue & profit drivers.
- Hidden Gems → High margin but low visibility.
- Loss Makers → Require corrective action.

Business Value: Supports inventory and marketing optimization.

Root Cause Analysis

Primary profitability drivers identified:

- Excessive discounting.
- Low-margin Furniture category.
- Regional underperformance (South).
- High sales concentration in limited SKUs.

Risk / Anomaly Analysis

- Transactions with high discount and negative profit flagged.
- Certain sub-categories consistently loss-making.
- Profit highly concentrated in few products (concentration risk).

Scenario Analysis

If discount is capped at 20%:

- Negative margin transactions reduce significantly.
- Overall margin improves.
- Revenue impact expected to be moderate.

Dashboard Design

The dashboard was developed in **Google Sheets** using pivot tables, formulas, and interactive slicers. It transforms transactional retail data into a structured profit intelligence interface for decision-making.

Dashboard Objective

The dashboard is designed to:

- Monitor core KPIs: **Total Sales, Total Profit, Orders, Profit Margin, YoY Growth**
- Identify profit erosion caused by discounting and low-margin categories
- Enable dynamic analysis across time, region, and product dimensions

Dashboard Structure

The dashboard is divided into the following sections:

1. Executive KPI View

Displays overall performance indicators for quick decision-making.

Total Sales	Total Profit	Total Orders	Average Profit Margin (%)	Year-over -Year Growth (%)
\$3,488,320.50	\$228,044.95	7632	6.54%	13.57%

2. Regional & Category Analysis

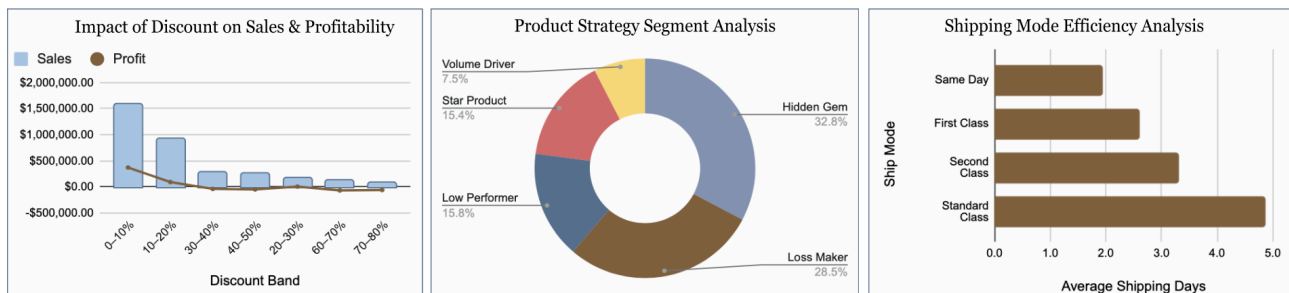
Compares sales and profitability across regions and product categories.



Explanation: Highlights regional imbalance and category-level margin differences.

3. Discount & Product Segmentation View

Analyzes the impact of discount levels and product performance classification.



Explanation: Demonstrates how higher discounts affect margins and shows product segmentation for strategic focus

Interactive Features

The dashboard includes dynamic filters (slicers) for:

- Year
- Region
- Category

All charts and KPIs update automatically based on selected filters, enabling drill-down analysis.

Conclusion

The dashboard converts raw transaction data into a profit-focused decision support system, allowing stakeholders to move from revenue tracking to margin optimization.

Insights Summary

Below are the most critical insights derived from the analysis, written in decision-oriented language:

1. **Revenue growth is not translating into strong profitability** (Overall margin $\approx$ 6.5%).
2. **23% of transactions have discounts above 20%**, significantly reducing margins.

3. **Profit margins decline sharply beyond the 20% discount threshold.**
4. **Furniture category operates at critically low margin (~0.74%),** with 44% of products loss-making.
5. **West region contributes 37% of total profit,** outperforming other regions.
6. **South region underperforms** in profitability relative to sales volume.
7. **3,157 transactions generate negative profit,** totaling ≈ \$330K in losses.
8. **33% of products are non-profitable,** indicating portfolio inefficiency.
9. **Top-performing products drive the majority of positive profit,** creating concentration risk.
10. **Sales are seasonally strong in September and November,** offering targeted planning opportunities.

Recommendations

Insight	Recommendation	Business Impact
High discount-driven losses	Cap discounts at 15–20% and introduce approval workflow	Protects margin, reduces negative-profit transactions
Furniture margin weakness	Reprice SKUs, renegotiate suppliers, remove loss-making items	Improves category profitability
Regional imbalance	Benchmark West practices; audit South pricing & logistics	Improves regional efficiency
High number of loss-making SKUs	Portfolio rationalization & focus on Star Products	Reduces risk, improves margin stability
Seasonal peaks	Align promotions with high-performing months	Maximizes revenue with controlled discounting
Revenue-focused KPI culture	Shift reporting emphasis toward Profit Margin & YoY Profit	Encourages sustainable decision-making

Impact Estimation

If implemented, the recommendations are expected to deliver:

Cost Savings

- Reducing negative-profit transactions (currently \$330K losses) could potentially recover **\$100K-\$150K annually through** discount control and SKU optimization.

Efficiency Improvement

- Portfolio rationalization (removing 33% non-profitable SKUs) reduces inventory complexity and holding costs.
- Regional benchmarking improves operational consistency.

Service Improvement

- Focused inventory on high-margin products ensures better stock availability.
- Targeted promotions improve customer satisfaction without excessive discounting.

Risk Reduction

- Lower dependence on a small set of products reduces profit concentration risk.
- Discount cap reduces volatility in margins.

Estimated overall margin improvement potential: **+2-4 percentage points**.

Limitations

Despite valuable findings, the analysis has certain limitations:

- Dataset limited to historical data (2014-2017).
- No detailed cost breakdown (COGS not available).
- No customer-level data for behavioral analysis.
- Scenario analysis (discount cap simulation) not quantitatively modeled.
- External factors (competition, macroeconomic conditions) not included.

Therefore, conclusions are based on internal transactional data only.

Future Scope

Further improvements can include:

- Predictive sales forecasting using time-series models.
- What-if simulation for discount optimization.
- Customer segmentation (RFM analysis) if customer data becomes available.
- Integration with real-time BI tools (Power BI/Tableau).
- Demand elasticity modeling for dynamic pricing strategy.

Additional data required:

- Cost of goods sold (COGS)
- Customer demographics & purchase history
- Marketing campaign data
- Inventory holding cost data

Conclusion

This project successfully transformed raw retail transaction data into a structured profit intelligence framework.

While the business generates strong revenue (\$3.48M), profitability (6.5% margin) is constrained by aggressive discounting, low-margin categories, and regional imbalance.

The dashboard and analytical framework provide actionable insights to shift the organization from revenue-focused decisions to sustainable profit optimization.

By implementing targeted discount control, category restructuring, and portfolio rationalization, the business can significantly improve margin stability, reduce risk, and enhance long-term financial performance.

Contribution Matrix

Team Member	Dataset & Sourcing	Cleaning	KPI & Analysis	Dashboard	Report Writing	PPT	Overall Role
Shubham Aggarwal	✓	✓	✓	✓	✓		Dashboard Lead & Strategy Lead
Kartik Mehra	✓		✓		✓		Data Lead
Priyanshu Agrahari	✓	✓	✓			✓	Project Lead & Analysis Lead
Prajwal Tripathi	✓					✓	PPT & Quality Lead
Mukund Mangla	✓					✓	
Jatin	✓						